HDOC Day 61 - Two Sum Using Pointers

Problem Statement

Write a C program to read the size and elements of a variable-length array of positive integers and a target value. Pass the array to a function named **two_sum()** using its address. Using nested loops, find two different indices whose elements add up to the target. Print the indices separated by a space, or print -1 -1 if no pair exists. At most one valid pair exists.

C Program

```
#include <stdio.h>

void two_sum(int *nums, int n, int target) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (*(nums + i) + *(nums + j) == target) {
                printf("%d %d\n", i, j);
                return;
            }
        }
    }
    printf("-1 -1\n");
}

int main(void) {
    int n, target;
    scanf("%d", &n);
    if (n <= 0) {
        printf("-1 -1\n");
        return 0;
    }

    int nums[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }
    scanf("%d", &target);

    two_sum(nums, n, target);
    return 0;
}
```

Sample Input

```
4
2 7 11 15
9
```

Sample Output

```
0 1
```

Explanation

The elements at indices 0 and 1 are 2 and 7. Their sum is 9, which matches the target. The function uses a pointer to access array elements and nested loops to check each distinct pair.

Time Complexity: $O(n^2)$ | Extra Space: O(1)